The Finance Trail

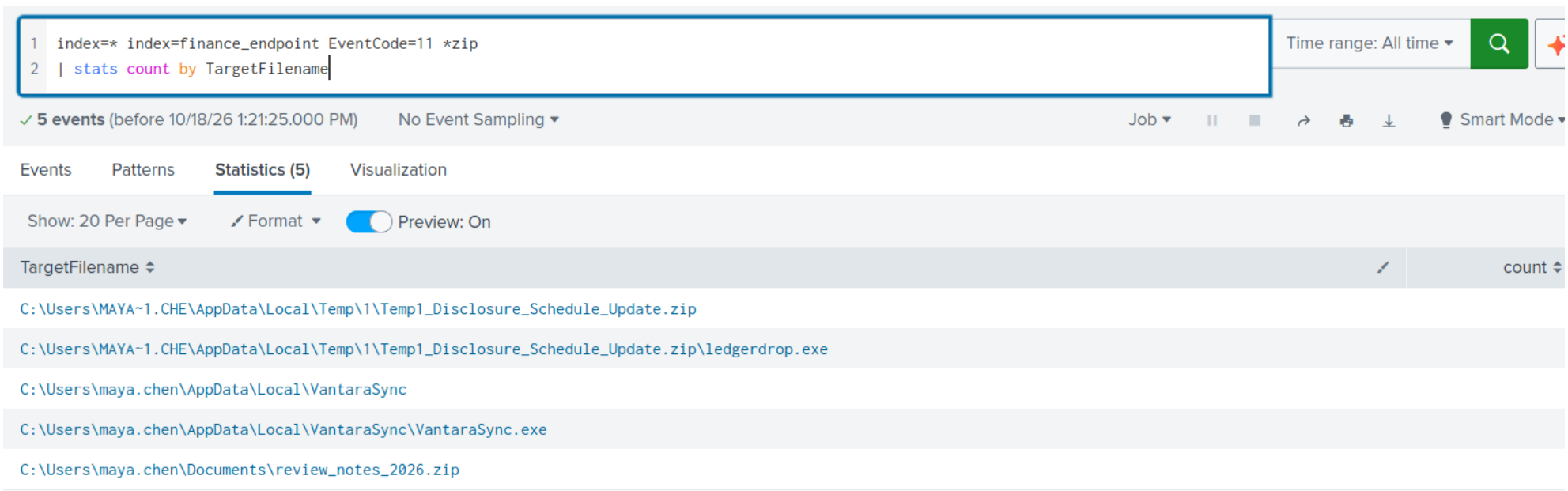
1. **The finance employee's workstation ran a program from inside a downloaded ZIP. Find the delivered email whose sender domain also hosted that ZIP. What is its Message-ID, without angle brackets?**
2. **After the lure email arrived, which parent executable launched the browser process that opened the external review site? Give the .exe filename.**
3. **On the affected workstation, what SHA-256 did Sysmon's file-stream hash event record when Chrome saved the ZIP that contained the executed loader?**
4. **What SHA-256 do the extracted loader and installed sync client share in their process-creation events?**
5. **Which scheduled task belongs to the loader's process lineage, not routine document syncing? Omit the leading backslash.**
6. **The sync client did not call out the moment it started. How many seconds passed between its Sysmon process-creation UtcTime and its first command-and-control (C2) callback in the Nginx web log, rounded to the nearest second?**
7. **For the loader-installed sync client's first successful callback from the affected workstation, which local TCP source port appears in both Suricata HTTP and the Windows Filtering Platform (WFP) allowed-connection event?**
8. **The main C2 server returned HTTP 503, and after that the sync client used a backup domain. In Suricata, the Domain Name System (DNS) answer of type A for the backup domain was recorded before that 503. How many milliseconds before the 503 was it recorded, to three decimal places?**
9. **Find the PowerShell process that created the archive later written to the share. In that workstation's captured Sysmon process-creation history, which executable last used its process ID before PowerShell started? Give the .exe filename.**
10. **The attacker staged an archive on the workstation and then moved it to the file server. How many whole seconds passed between the archive's Sysmon file-creation UtcTime and its successful SMB2 WRITE in Suricata?**

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1. The finance employee's workstation ran a program from inside a downloaded ZIP. Find the delivered email whose sender domain also hosted that ZIP. What is its Message-ID, without angle brackets?

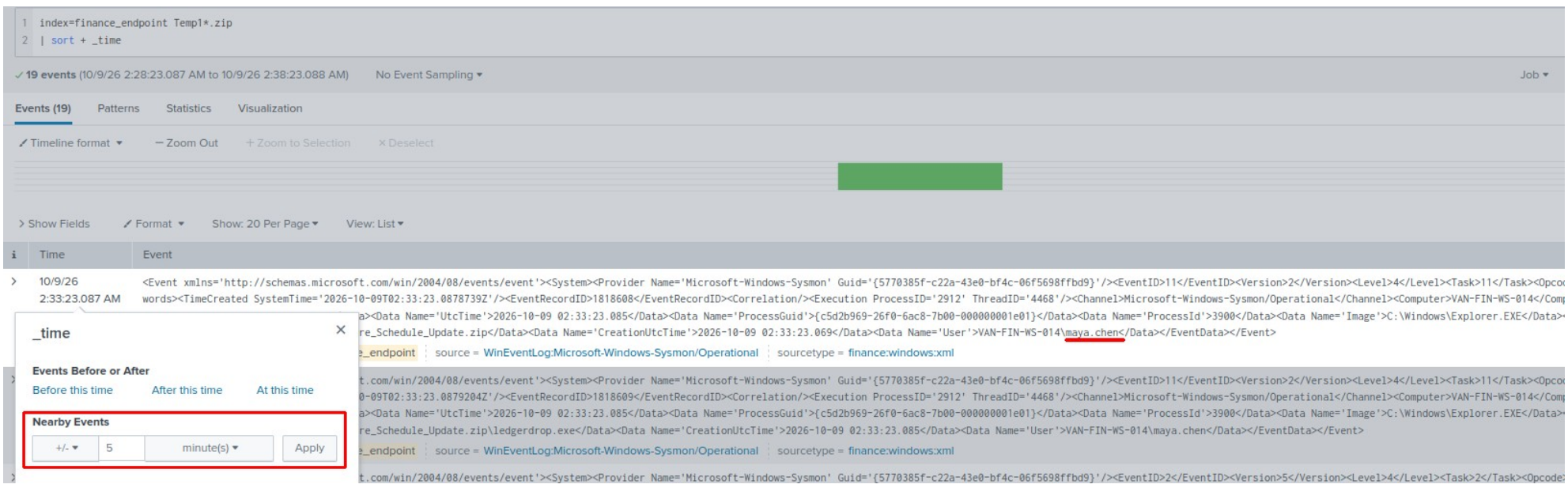
First, I needed to find the ZIP itself, so I searched for file creation events (EventCode 11) with the zip extension

Query: *index=* index=finance_endpoint EventCode=11 *zip*
| stats count by TargetFilename



We can see `ledgerdrop.exe` inside `Temp1_Disclosure_Schedule_Update.zip`. Windows Explorer creates the `Temp1_<name>.zip` folder in Temp when a user runs a file directly from an opened archive, so this is the program from the question. Next, I opened these events: the user is `maya.chen` on `VAN-FIN-WS-014`, and the program was run at 02:33:23. To see what happened before that, I used Nearby Events (+/- 5 minutes)

Query: *index=finance_endpoint Temp1*.zip*
| sort + _time

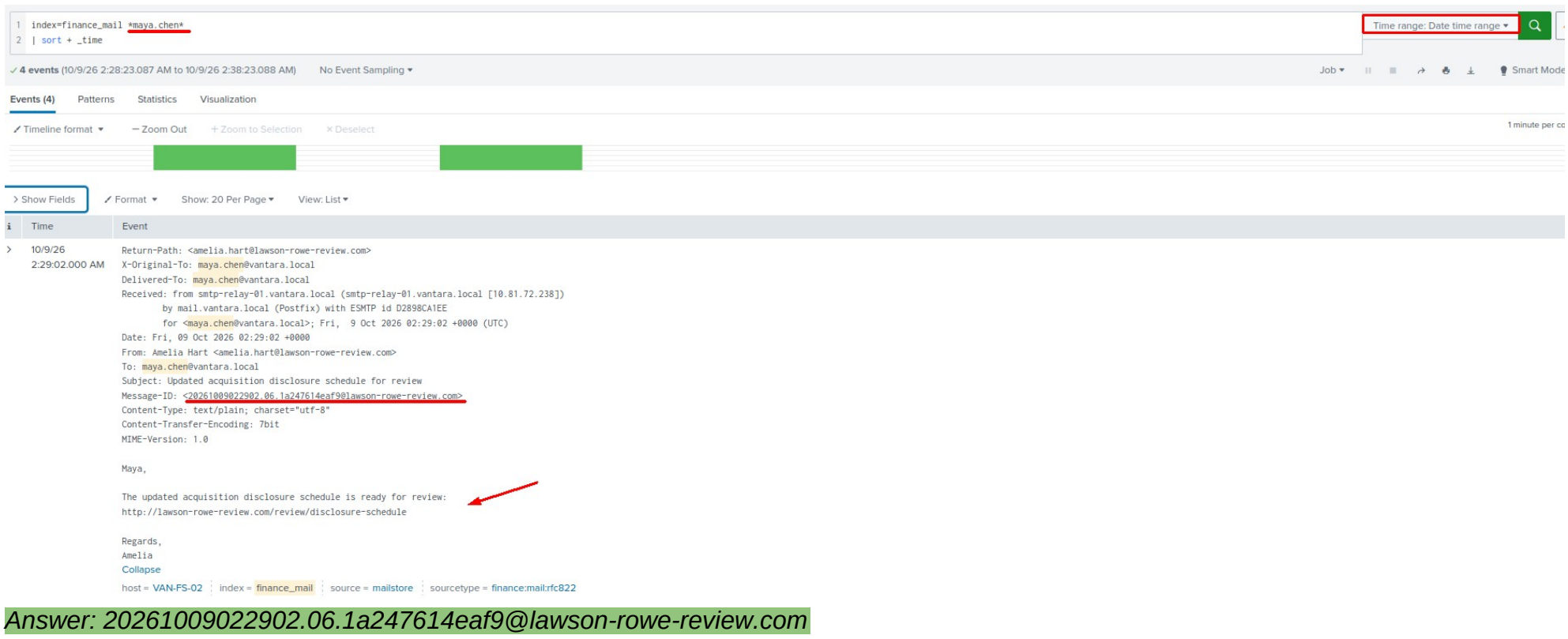


With the same time range, I searched the mail index for `maya.chen`. At 02:29:02, she received an email from `amelia.hart@lawson-rowe-review.com` with the subject \"Updated acquisition disclosure schedule for review\" and the link:

```
http://lawson-rowe-review.com/review/disclosure-schedule
```

The link points to the sender's own domain, and a few minutes after this email Chrome downloaded the ZIP (see Q2 and Q3). So `lawson-rowe-review.com` is the domain that both sent the email and hosted the ZIP

Query: *index=finance_mail *maya.chen**
| sort + _time

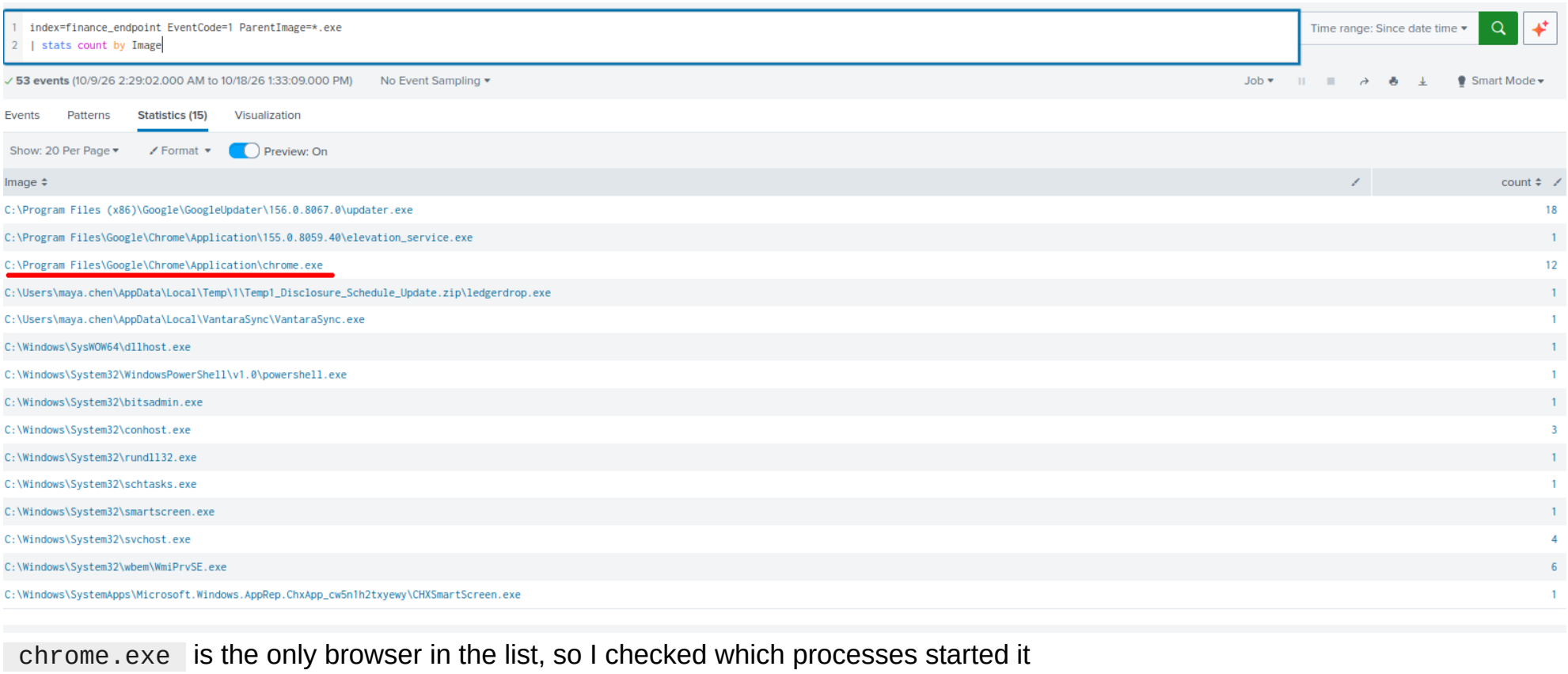


Answer: 20261009022902.06.1a247614eaf9@lawson-rowe-review.com

2. **After the lure email arrived, which parent executable launched the browser process that opened the external review site? Give the .exe filename.**

The email arrived at 02:29:02, so I set the time range to start from this moment and checked all processes started by another exe file

Query: *index=finance_endpoint EventCode=1 ParentImage=*.exe*
| stats count by Image



`chrome.exe` is the only browser in the list, so I checked which processes started it

Query: *index=finance_endpoint EventCode=1 ParentImage=*.exe Image="C:\\Program Files\\Google\\Chrome\\Application\\chrome.exe"*
| stats count by ParentImage



11 of 12 Chrome processes were started by `chrome.exe` itself (these are its child processes: tabs, GPU, etc.). Only one was started by another program:

```
C:\Program Files\Vantara\Finance Client\VantaraFinancePortal.exe
```

This is the process that opened the link to the external review site, after that Chrome only created its own child processes

Answer: VantaraFinancePortal.exe

3. **On the affected workstation, what SHA-256 did Sysmon's file-stream hash event record when Chrome saved the ZIP that contained the executed loader?**

When a browser downloads a file, it adds the `Zone.Identifier` alternate data stream to it (Mark of the Web). Sysmon logs this as EventCode 15 (FileCreateStreamHash) together with the hash of the file. So I checked these events after the email arrived

Query: *index=* EventCode=15*
| table Image, TargetFilename, Hash

```
1 index=* EventCode=15
2 | table Image, TargetFilename, Hash
```

Time range: Since date time

✓ **6 events** (10/9/26 2:29:02.000 AM to 10/18/26 1:41:19.000 PM) No Event Sampling — Job — Smart Mode

Events | Patterns | **Statistics (6)** | Visualization

Show: 20 Per Page — Format — Preview: On

Image	TargetFilename	Hash
C:\Users\maya.chen\AppData\Local\Temp\1\Temp1_Disclosure_Schedule_Update.zip\ledgerdrop.exe	C:\Users\maya.chen\AppData\Local\VantaraSync\VantaraSync.exe	SHA256=AAD4B231C4AFC7D02A09344545A957086FE36CD6A9CCA8F1F92E1AC3DDFFAC5C
C:\Windows\Explorer.EXE	C:\Users\maya.chen\AppData\Local\Temp\1\Temp1_Disclosure_Schedule_Update.zip\ledgerdrop.exe	SHA256=AAD4B231C4AFC7D02A09344545A957086FE36CD6A9CCA8F1F92E1AC3DDFFAC5C
C:\Windows\Explorer.EXE	C:\Users\maya.chen\AppData\Local\Temp\1\Temp1_Disclosure_Schedule_Update.zip\ledgerdrop.exe	SHA256=AAD4B231C4AFC7D02A09344545A957086FE36CD6A9CCA8F1F92E1AC3DDFFAC5C
C:\Program Files\Google\Chrome\Application\chrome.exe	C:\Users\maya.chen\Downloads\Disclosure_Schedule_Update.zip	SHA256=98D66005B3F2CFD556EEF503EC1ADA73DE34C24D3800D41D8E30C43AF32DEDEF
C:\Program Files\Google\Chrome\Application\chrome.exe	C:\Users\maya.chen\Downloads\Disclosure_Schedule_Update.zip	SHA256=98D66005B3F2CFD556EEF503EC1ADA73DE34C24D3800D41D8E30C43AF32DEDEF
C:\Program Files\Google\Chrome\Application\chrome.exe	C:\Users\maya.chen\Downloads\Disclosure_Schedule_Update.zip	SHA256=98D66005B3F2CFD556EEF503EC1ADA73DE34C24D3800D41D8E30C43AF32DEDEF

`chrome.exe` saved `C:\Users\maya.chen\Downloads\Disclosure_Schedule_Update.zip`. This is the same archive that was later opened as `Temp1_Disclosure_Schedule_Update.zip`, and the loader `ledgerdrop.exe` was run from it (Q1)

Answer: 98D66005B3F2CFD556EEF503EC1ADA73DE34C24D3800D41D8E30C43AF32DEDEF

4. **What SHA-256 do the extracted loader and installed sync client share in their process-creation events?**

In the same results as in Q3, there are two more files with the same hash:

```
Explorer.EXE   -> ...\Temp1_Disclosure_Schedule_Update.zip\ledgerdrop.exe
ledgerdrop.exe -> C:\Users\maya.chen\AppData\Local\VantaraSync\VantaraSync.exe
```

Explorer extracted the loader `ledgerdrop.exe`, and then the loader installed the sync client `VantaraSync.exe`. Both files have the same SHA-256, which means that the loader simply copied itself under a name that looks like a legitimate Vantara application. These are the files that were executed later (Q1 and Q6), so their process creation events contain the same hash

Query: *index=* EventCode=15*
| table Image, TargetFilename, Hash

```
1 index=* EventCode=15
2 | table Image, TargetFilename, Hash
```

Time range: Since date time

✓ **6 events** (10/9/26 2:29:02.000 AM to 10/18/26 1:41:19.000 PM) No Event Sampling — Job — Smart Mode

Events | Patterns | **Statistics (6)** | Visualization

Show: 20 Per Page — Format — Preview: On

Image	TargetFilename	Hash
C:\Users\maya.chen\AppData\Local\Temp\1\Temp1_Disclosure_Schedule_Update.zip\ledgerdrop.exe	C:\Users\maya.chen\AppData\Local\VantaraSync\VantaraSync.exe	SHA256=AAD4B231C4AFC7D02A09344545A957086FE36CD6A9CCA8F1F92E1AC3DDFFAC5C
C:\Windows\Explorer.EXE	C:\Users\maya.chen\AppData\Local\Temp\1\Temp1_Disclosure_Schedule_Update.zip\ledgerdrop.exe	SHA256=AAD4B231C4AFC7D02A09344545A957086FE36CD6A9CCA8F1F92E1AC3DDFFAC5C
C:\Windows\Explorer.EXE	C:\Users\maya.chen\AppData\Local\Temp\1\Temp1_Disclosure_Schedule_Update.zip\ledgerdrop.exe	SHA256=AAD4B231C4AFC7D02A09344545A957086FE36CD6A9CCA8F1F92E1AC3DDFFAC5C
C:\Program Files\Google\Chrome\Application\chrome.exe	C:\Users\maya.chen\Downloads\Disclosure_Schedule_Update.zip	SHA256=98D66005B3F2CFD556EEF503EC1ADA73DE34C24D3800D41D8E30C43AF32DEDEF
C:\Program Files\Google\Chrome\Application\chrome.exe	C:\Users\maya.chen\Downloads\Disclosure_Schedule_Update.zip	SHA256=98D66005B3F2CFD556EEF503EC1ADA73DE34C24D3800D41D8E30C43AF32DEDEF
C:\Program Files\Google\Chrome\Application\chrome.exe	C:\Users\maya.chen\Downloads\Disclosure_Schedule_Update.zip	SHA256=98D66005B3F2CFD556EEF503EC1ADA73DE34C24D3800D41D8E30C43AF32DEDEF

Answer: AAD4B231C4AFC7D02A09344545A957086FE36CD6A9CCA8F1F92E1AC3DDFFAC5C

5. **Which scheduled task belongs to the loader's process lineage, not routine document syncing? Omit the leading backslash.**

To find the scheduled task, I searched for EventCode 4698 (A scheduled task was created) with the names of both malicious files

Query: *index=* EventCode=4698 (*VantaraSync* OR *ledgerdrop*)*
| table TaskName

New Search

```
1 index=* EventCode=4698 (*VantaraSync* OR *ledgerdrop*)
2 | table TaskName
```

✓ **1 event** (10/9/26 2:29:02.000 AM to 10/18/26 1:50:10.000 PM) No Event Sampling

Events | Patterns | **Statistics (1)** | Visualization

Show: 20 Per Page — Format — Preview: On

TaskName
\VantaraUpdateCheck

To be sure that this task belongs to the loader and is not a normal sync task, I checked its process lineage. In the screenshot for Q6, `schtasks.exe` that created `VantaraUpdateCheck` has `ledgerdrop.exe` as its parent process:

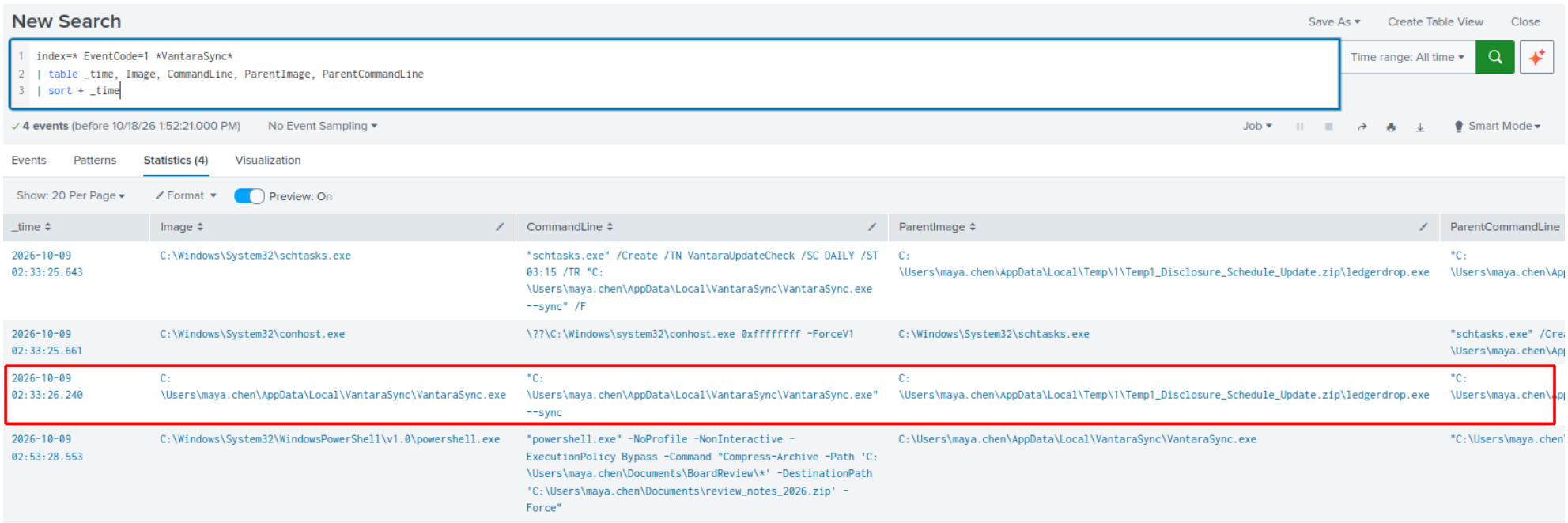
```
"schtasks.exe" /Create /TN VantaraUpdateCheck /SC DAILY /ST 03:15
  /TR "C:\Users\maya.chen\AppData\Local\VantaraSync\VantaraSync.exe --sync" /F
ParentImage: ...\Temp1_Disclosure_Schedule_Update.zip\ledgerdrop.exe
```

Answer: VantaraUpdateCheck

6. **The sync client did not call out the moment it started. How many seconds passed between its Sysmon process-creation UtcTime and its first command-and-control (C2) callback in the Nginx web log, rounded to the nearest second?**

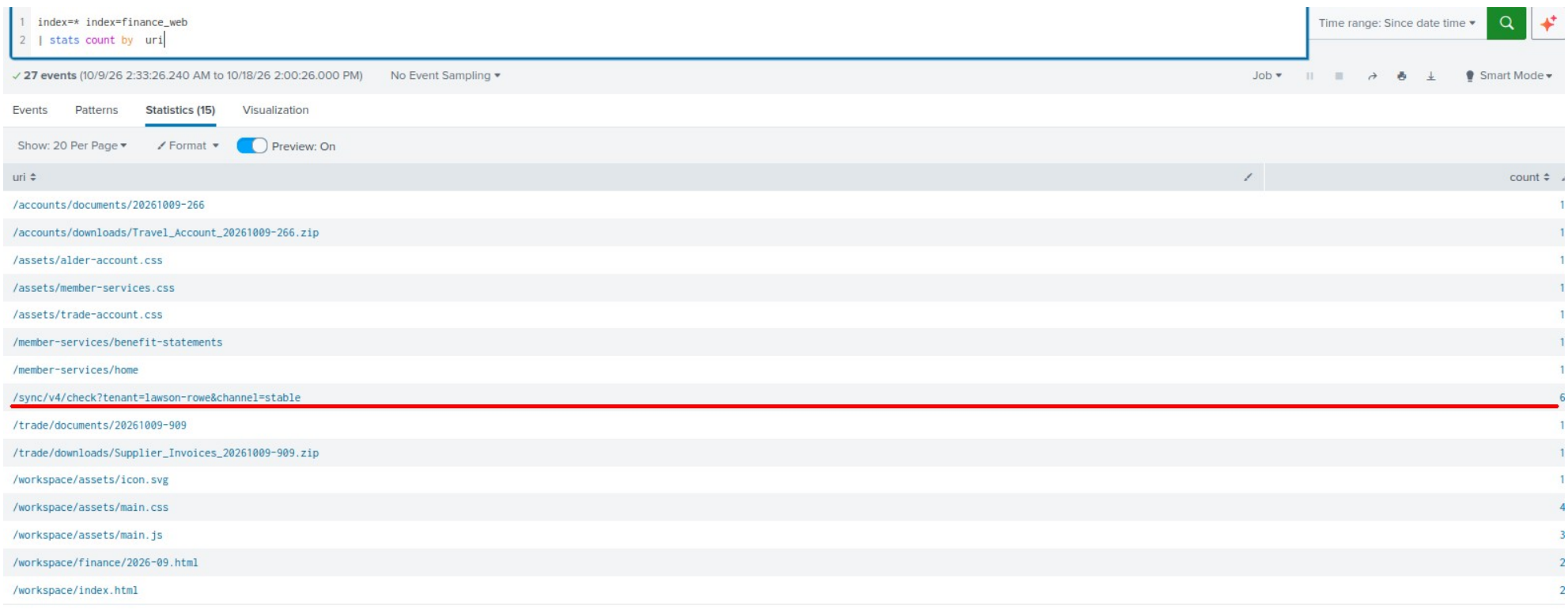
First, I checked all process creation events related to VantaraSync and sorted them by time

Query: *index=* EventCode=1 *VantaraSync**
| table _time, Image, CommandLine, ParentImage, ParentCommandLine
| sort + _time



`ledgerdrop.exe` started `VantaraSync.exe --sync` at 2026-10-09 02:33:26.240. Then I moved to the Nginx web logs and checked all URIs starting from this time. One URI was requested 6 times, which looks like a beacon

Query: *index=* index=finance_web*
| stats count by uri



I sorted these requests by time. The first one has the user-agent `VantaraSync/4.1`, comes from the workstation `10.81.70.254` and goes to `cdn-lawsonreview.net` (`203.0.113.45`). The response code 204 (No Content) is typical for a C2 check-in without tasks. The time of this first callback is 02:34:13

Query: *index=* index=finance_web uri="/sync/v4/check?tenant=lawson-rowe&channel=stable"*
| sort + _time

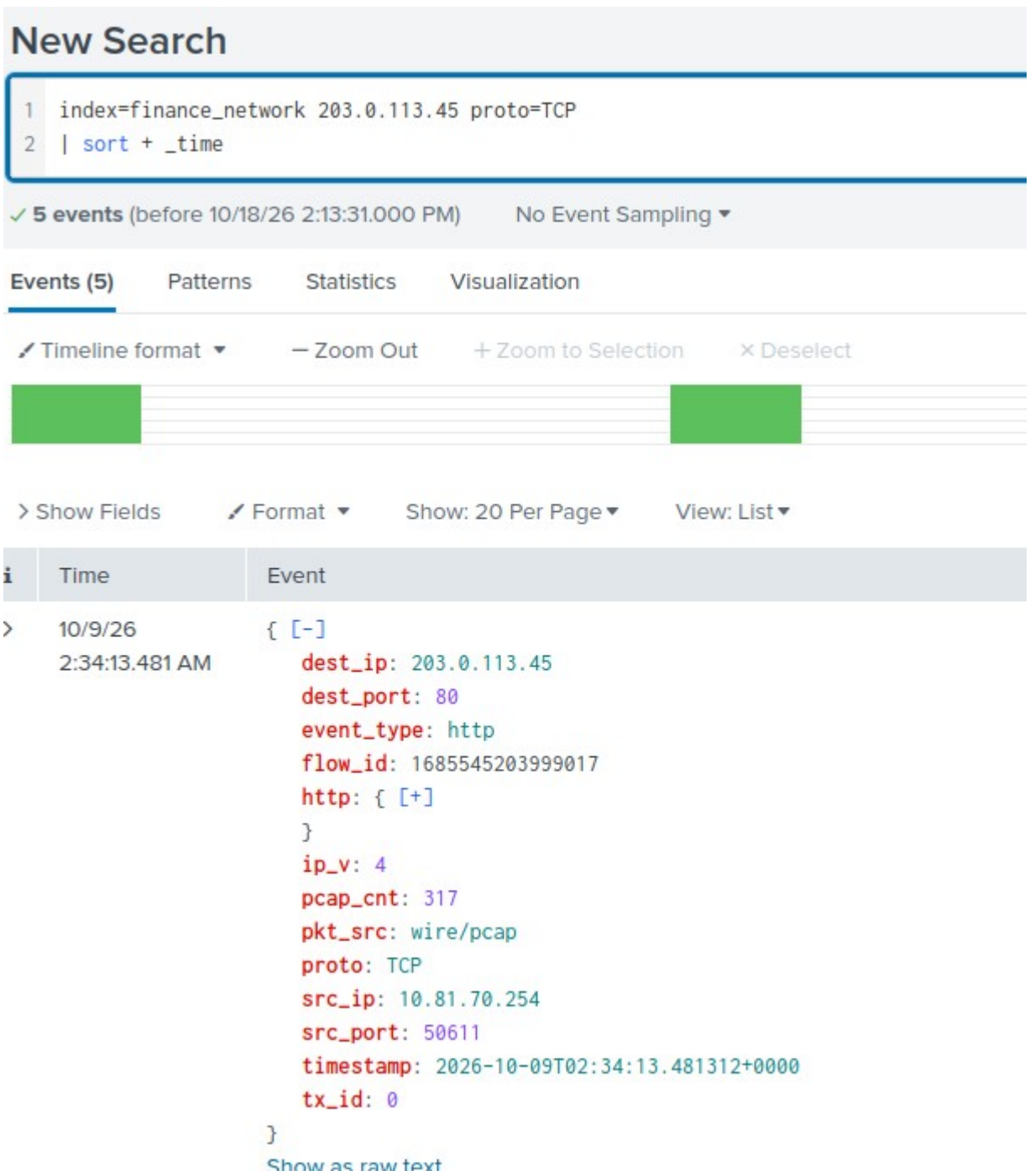
02:34:13 - 02:33:26.240 = 46.76 seconds, which is 47 after rounding

Answer: 47

7. **For the loader-installed sync client's first successful callback from the affected workstation, which local TCP source port appears in both Suricata HTTP and the Windows Filtering Platform (WFP) allowed-connection event?**

We know the C2 IP address from Q6, so I searched for TCP connections to it in the Suricata logs and sorted them by time

Query: *index=finance_network 203.0.113.45 proto=TCP*
| sort + _time



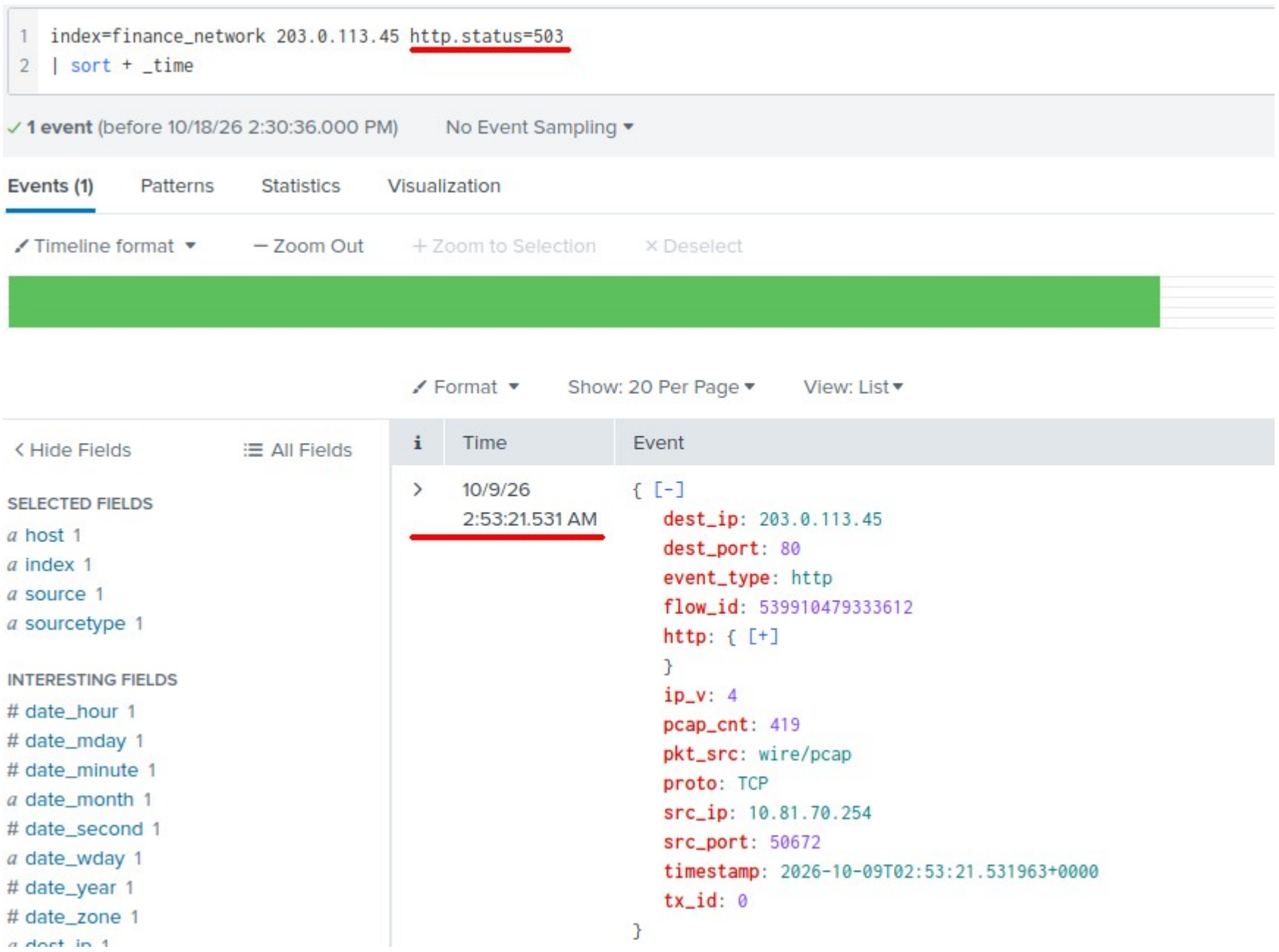
The first event is an HTTP flow from the workstation `10.81.70.254` to `203.0.113.45:80` at 02:34:13.481. This is the same second as the first callback in the Nginx log from Q6, so it is the same connection. Its source port is `50611`. According to the question, the same port also appears in the WFP allowed-connection event (EventCode 5156)

Answer: 50611

8. **The main C2 server returned HTTP 503, and after that the sync client used a backup domain. In Suricata, the Domain Name System (DNS) answer of type A for the backup domain was recorded before that 503. How many milliseconds before the 503 was it recorded, to three decimal places?**

To find the 503, I filtered the Suricata logs by the C2 IP address and the HTTP 503 (Service Unavailable) status code

Query: *index=finance_network 203.0.113.45 http.status=503*
| sort + _time



There is only one such event, and its timestamp is 2026-10-09T02:53:21.531963. Then I checked the Suricata DNS events around this time and found the type A answer for the backup domain `api-lawsoncdn.net` at 2026-10-09T02:53:21.523656

21.531963 - 21.523656 = 0.008307 seconds = 8.307 milliseconds
Answer: 8.307

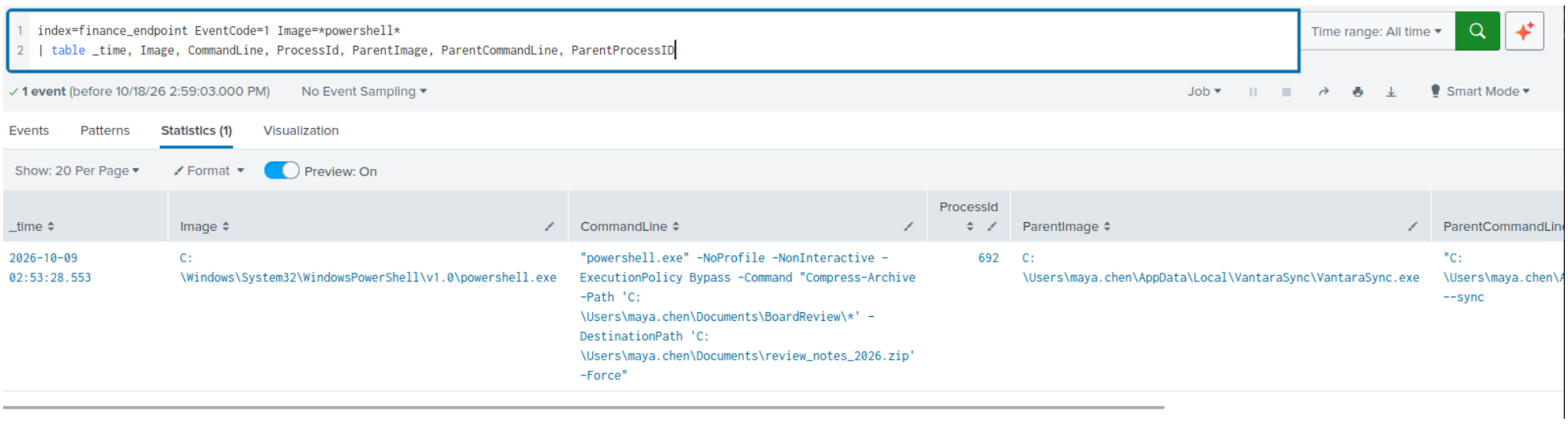
9. **Find the PowerShell process that created the archive later written to the share. In that workstation's captured Sysmon process-creation history, which executable last used its process ID before PowerShell started? Give the .exe filename.**

In the screenshot from Q6, we saw that `VantaraSync.exe` started PowerShell, which archives the BoardReview folder into `review_notes_2026.zip` (this archive is later written to the share in Q10):

```
powershell.exe -NoProfile -NonInteractive -ExecutionPolicy Bypass
  -Command "Compress-Archive -Path 'C:\Users\maya.chen\Documents\BoardReview\*'
  -DestinationPath 'C:\Users\maya.chen\Documents\review_notes_2026.zip' -Force"
```

I checked this event in more detail to get its `ProcessId`

Query: *index=finance_endpoint EventCode=1 Image=*powershell**
| table _time, Image, CommandLine, ProcessId, ParentImage, ParentCommandLine, ParentProcessID



The PID is 692. Windows reuses PIDs after processes finish, so I searched for all process creation events with the same PID

Query: *index=finance_endpoint EventCode=1 ProcessId=692*
| table _time, Image, CommandLine, ProcessId, ParentImage, ParentCommandLine, ParentProcessID

```
1 index=finance_endpoint EventCode=1 ProcessId=692
2 | table _time, Image, CommandLine, ProcessId, ParentImage, ParentCommandLine, ParentProcessID
```

Time range: All time

✓ **2 events** (before 10/18/26 2:59:33.000 PM) No Event Sampling — Job — Smart Mode

Events | Patterns | **Statistics (2)** | Visualization

Show: 20 Per Page — Format — Preview: On

_time	Image	CommandLine	ProcessId	ParentImage
2026-10-09 02:53:28.553	C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe	"powershell.exe" -NoProfile -NonInteractive -ExecutionPolicy Bypass -Command "Compress-Archive -Path 'C:\Users\maya.chen\Documents\BoardReview*' -DestinationPath 'C:\Users\maya.chen\Documents\review_notes_2026.zip' -Force"	692	C:\Users\maya.chen\AppData\Local\VantaraSync\Vanta
2026-10-09 02:33:14.259	C:\Program Files\Google\Chrome\Application\155.0.8059.40\elevation_service.exe	"C:\Program Files\Google\Chrome\Application\155.0.8059.40\elevation_service.exe"	692	C:\Windows\System32\services.exe

There are only two events. Before PowerShell (02:53:28), PID 692 was used by Chrome's `elevation_service.exe` at 02:33:14
Answer: elevation_service.exe

10. The attacker staged an archive on the workstation and then moved it to the file server. How many whole seconds passed between the archive's Sysmon file-creation UtcTime and its successful SMB2 WRITE in Suricata?

First, I checked SMB2 WRITE operations in Suricata. The first one writes `review_notes_2026.zip` from the workstation `10.81.70.254` to the file server `10.81.72.63` (port 445) with `STATUS_SUCCESS` at 02:53:40.986626

Query: *index=finance_network event_type=smb smb.command=SMB2_COMMAND_WRITE*

```
1 index=finance_network  event_type=smb smb.command=SMB2_COMMAND_WRITE
```

✓ **3 events** (before 10/18/26 2:58:04.000 PM) No Event Sampling

Events (3) | Patterns | Statistics | Visualization

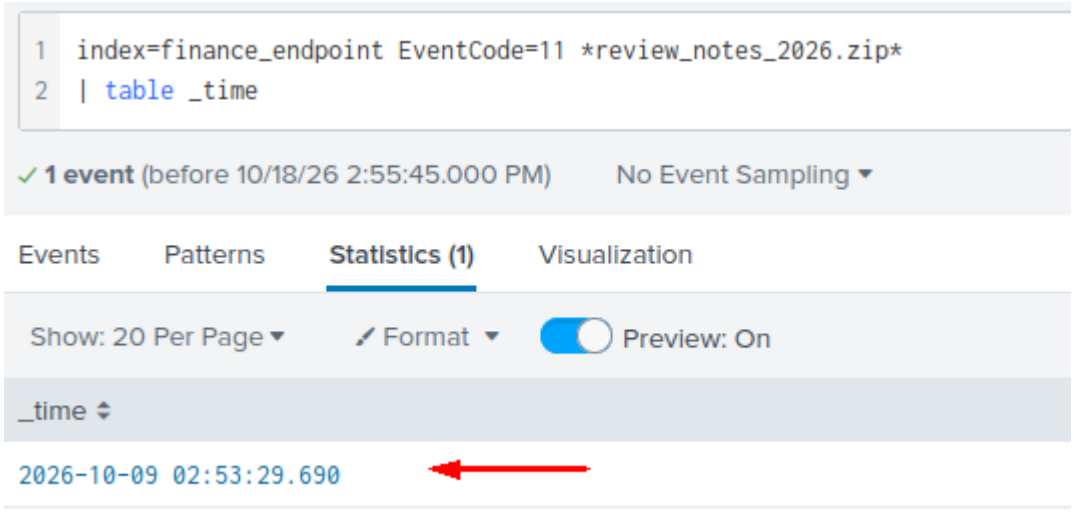
Timeline format — Zoom Out — + Zoom to Selection — × Deselect

> Show Fields — Format — Show: 20 Per Page — View: List

i	Time	Event
>	10/9/26 2:53:40.986 AM	{ [-] dest_ip: 10.81.72.63 dest_port: 445 event_type: smb flow_id: 1289574328462318 ip_v: 4 pcap_cnt: 467 pkt_src: wire/pcap proto: TCP smb: { [-] command: SMB2_COMMAND_WRITE dialect: 3.11 filename: review_notes_2026.zip fuid: dd87e230-0000-0000-930b-177100000000 id: 11 session_id: 1238275884 share: status: STATUS_SUCCESS status_code: 0x0 tree_id: 2766169606 } src_ip: 10.81.70.254 src_port: 50677 timestamp: 2026-10-09T02:53:40.986626+0000

Then I found the file creation event (EventCode 11) for this archive on the workstation. It was created at 02:53:29.690, one second after PowerShell from Q9 started

Query: *index=finance_endpoint EventCode=11 *review_notes_2026.zip**
| table _time



02:53:40.986626 - 02:53:29.690 = 11.296626 seconds, so 11 whole seconds
Answer: 11